PART-3

1. Data Collection

Data is collected in a structured format using Python libraries like Pandas. In this assignment, the dataset contains Study_Hours, Attendance, Assignments, and Marks. This data is used for building a machine learning model.

2. Data Cleaning

Data cleaning involves checking for missing values, incorrect data, and duplicates. In this dataset, the data is already clean, so no major cleaning steps are required.

3. Feature Selection

Feature selection means choosing important input variables for prediction. Here, Study_Hours, Attendance, and Assignments are selected as features (X). Marks is selected as the target variable (y).

4. Train-Test Split

The dataset is divided into training and testing data. Usually 80% is used for training and 20% for testing. Training data helps the model learn patterns, and testing data checks model performance.

5. Model Training

A Linear Regression model is used in this assignment. The model is trained using training data to learn the relationship between features and target variable.

6. Testing

After training, the model is tested using unseen test data. This helps to check how well the model performs on new data.

7. Evaluation

Model performance is evaluated using metrics like Mean Absolute Error (MAE) and R^2 Score. These metrics show how accurate the model predictions are.

8. Prediction

The trained model is used to predict marks for new input values. It uses learned patterns to estimate output.